Hyemin Gu

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Amherst, MA 01002, USA

Research Interest

Mathematically Grounded Machine Learning; Generative Modeling; Stable Learning Algorithms; Dynamical Systems; Scientific Applications of Machine Learning

Employment

- **University of Massachusetts Amherst** *Oct 2025 – Present*
  Postdoctoral Researcher Amherst, MA
  - Advisor: Markos A. Katsoulakis
  - Developing mathematically grounded machine learning architectures for complex learning and optimization, and dynamical system simulators for supply-chain network systems under DARPA's The Right Space Program.
  - Leading the mathematical formulation and algorithmic development of gradient flow-based generative models for continuous prediction of single-cell dynamics.
- **Ewha Womans University Seoul Hospital** *Jul 2020 – Dec 2020*
  Statistics Specialist Seoul, South Korea
  - Developed an R-based Bioconductor pipeline (e.g., `limma`, `edgeR`, `TCGAbiolinks`) for gene expression analysis, including survival analysis and pathway enrichment
  - Authored a tutorial book on reproducible gene expression workflows for biomedical researchers [R/Bioconductor tutorial]
  - Conducted statistical training sessions for clinicians, enhancing research reproducibility and data literacy

Education

- **University of Massachusetts Amherst** *Sep 2020 - Feb 2026*
  Ph.D. in Applied Mathematics Amherst, MA
  - Advisor: Markos A. Katsoulakis
  - Dissertation: Robust Generative Modeling with Wasserstein Proximal Regularization
- **Ewha Womans University** *Mar 2018 - Feb 2020*
  M.S. in Applied Mathematics Seoul, South Korea
  - Thesis: Convolutional Neural Network for 2D Flow Estimation Problem
- **Ewha Womans University** *Mar 2014 - Feb 2018*
  B.S. in Mathematics and Computational science, Minor in Statistics Seoul, South Korea
  - Thesis: Low cost training of a classification Neural Network with respect to Weight Selection

Publications

J=Journal, P=Preprint

[P.4] T Gamage, H Gu, Z Zhang, Z Chen, MA. Katsoulakis, L Rey-Bellet. (2026). **Fine-Tuning Generative Models for Extreme Events via CVaR-Penalized Wasserstein Gradient Flows** .

[P.3] H Gu, M Tyrrell, T Sahai, MA. Katsoulakis. (2026). **Sharp Stability Threshold and Certification for Designing Stable Residual Architectures**.

[J.3] YC Cheng, H Gu, et al. (2026). **PROFET Predicts Continuous Gene Expression Dynamics from scRNA-seq Data to Elucidate Heterogeneity of Cancer Treatment Responses**. *Cell Systems*, accepted.

[P.2] Z Zhang, H Gu, MA. Katsoulakis, T Sahai, et al. (2026). **ISOMORPH: A Supply Chain Digital Twin for Simulation, Dataset Generation, and Forecasting Benchmarks**.

[J.2] Z Chen, H Gu, MA Katsoulakis, L Rey-Bellet, W Zhu. (2025). **Robust Generative Learning with Lipschitz-Regularized α-Divergences Allows Minimal Assumptions on Target Distributions**. *Information and Inference: A Journal of the IMA*, Volume 14, Issue 4, December 2025, iaaf028. DOI: https://doi.org/10.1093/imaiai/iaaf028

[P.1] H Gu, MA Katsoulakis, L Rey-Bellet, BJ Zhang. (2024). **Combining Wasserstein-1 and Wasserstein-2 proximals: robust manifold learning via well-posed generative flows**.

[J.1] H Gu, P Birmpa, Y Pantazis, L Rey-Bellet, MA Katsoulakis. (2024). **Lipschitz-Regularized Gradient Flows and Generative Particle Algorithms for High-Dimensional Scarce Data**. *SIAM Journal of Mathematics of Data Science*, 6 (4), 1205-1235. DOI: https://doi.org/10.1137/23M1587841

Research Experience

- **University of Massachusetts Amherst** — *Oct 2025 – Present*
  Postdoctoral Research — Amherst, MA
  - Primary Research: **DARPA *The Right Space* (TRS) Program**

    TRAGOS: Developing (i) machine learning architectures for complex learning and optimization problems through principled mathematical formulations and (ii) dynamical system simulators for supply-chain network systems.
    - Developed a principled methodology for constructing stable residual neural architectures by establishing necessary and sufficient conditions through optimal-control and Hamilton–Jacobi formulations.
    - Developed ISOMORPH, a Markov-chain-based digital twin of a multi-echelon logistics network for scalable simulation and benchmark dataset generation, and built an interactive demonstration platform for forecasting research.
  - Major Collaboration: **Gradient Flow-Based Generative Modeling for Single-Cell Dynamics**
    - Led the mathematical and algorithmic development of PROFET, a gradient flow-based generative framework for continuous trajectory inference from sparse scRNA-seq data, achieving a 2.6–12.5× reduction in prediction error compared with 10 state-of-the-art methods.
  - Research Collaboration: **Sequential Refinement for Learning Heavy-Tailed Distributions**
    - Contributed to the algorithmic development by proposing the initial algorithmic implementation, identifying the CVaR-difference refinement strategy, and refining the algorithm to improve robustness.
- **University of Massachusetts Amherst** — *Feb 2022 – Sep 2025*
  Doctoral Research — Amherst, MA

- Primary Research: **Robust Generative Modeling with Wasserstein Proximal Regularization**
  - Developed gradient flow-based generative algorithms based on sequential particle transport using Wasserstein-1-regularized f-divergences, and extended the framework to latent-space dynamics for scalable high-dimensional learning.
  - Developed well-posed flow-based generative models using Wasserstein-1 and Wasserstein-2 proximal-regularized f-divergences for robust learning of distributions supported on low-dimensional manifolds.
- Research Validation: **Cross-Architecture Benchmarking of Wasserstein-Proximal Regularization**
  - Implemented and benchmarked Wasserstein-1 and Wasserstein-2 proximal regularization across diffusion models, normalizing flows, and GANs, demonstrating the data-agnostic robustness of Wasserstein-1 proximal regularization.
- Research Collaboration: **Gradient Flow-Based Generative Modeling for Single-Cell Dynamics**
  - Developed the mathematical formulation and core algorithmic framework of PROFET, a gradient flow-based generative model for continuous trajectory inference from sparse scRNA-seq data.

Invited Talks

- **Robust Mamba-Based TSF via Decomposed Lipschitz Regularization** at *SIAM Conference on Uncertainty Quantification (UQ)* in Minneapolis, MN, Mar 2026
- **Generative Particle Flows with Force-Matching for Reconstructing Nonlinear Cellular Dynamics** at *SIAM NNP Sectional Annual Meeting* in University Park, PA, Nov 2025
- **Combining Wasserstein-1 and Wasserstein-2 proximals: robust manifold learning via well-posed generative flows** at *Joint Mathematics Meetings – MAA* in Seattle, WA, Jan 2025
- **Wasserstein Proximals Stabilize Training of Generative Models and Learn Manifolds** at *Mathematics of Data Science – SIAM* in Atlanta, GA, Oct 2024

Poster Presentations

- **Data-Driven Force-Matching and Generative Particle Flows for Reconstructing Nonlinear Dynamics from Sparse scRNA-Seq Time Series** at *Sampling, Inference, and Data-Driven Physical Modeling in Scientific Machine Learning – IPAM* in Los Angeles, CA, Jul 2025
- **Lipschitz Regularized Gradient Flows and Latent Generative Particles** at *Mathematics of Data Science – SIAM* in Atlanta, GA, Oct 2024
- **Combining Wasserstein-1 and Wasserstein-2 proximals: robust manifold learning via well-posed generative flows** at *Robust Optimization and Simulation of Complex Stochastic Systems – ICERM* in Providence, RI, Sep 2024
- **Lipschitz Regularized Gradient Flows and Latent Generative Particles** at *Optimal Transport in Data Science – ICERM* in Providence, RI, May 2023
- **Training a 2 layer Neural Network using SVD-generated weights** at *Joint Mathematics Meetings – MAA* in San Diego, CA, Jan 2018
- **Necessary and sufficient conditions for shortest vectors in lattices of dimension 2 and 3** at *Joint Mathematics Meetings – MAA* in Atlanta, GA, Jan 2017

Honors and Awards

- **Distinguished Thesis Award** *Apr 2026*
  Department of Mathematics and Statistics, University of Massachusetts
- **Graduate Student Travel Grants** *Jan 2025*
  American Mathematical Society
- **Anne and Peter Costa Graduate Prize in Applied Mathematics** *Apr 2024*
  Department of Mathematics and Statistics, University of Massachusetts

Professional Service

- **Journal Reviewer**. *SIAM Journal on Scientific Computing (SISC)*, 2026

Teaching Experience

- **University of Massachusetts Amherst** *Feb 2021 – Dec 2024*
  Graduate Teaching Assistant Amherst, MA
  - (MATH456) **Mathematical Modeling – Mathematics of Generative AIs**: Graded assignments for an advanced modeling course focusing on generative AI methods. (Fall 2024)
  - (MATH532H) **Nonlinear Dynamics and Chaos**: Graded assignments and led Python tutorials for ODE solving. (Fall 2021)
  - (MATH545) **Linear Algebra for Applied Mathematics**: Graded assignments, held office hours, and conducted discussion sessions. (Summer 2021)
  - (MATH235) **Linear Algebra**: Graded assignments. (Spring 2021)

- **Ewha Womans University** *Mar 2018 – Dec 2019*
  Graduate Teaching Assistant Seoul, South Korea
  - **Numerical Differential Equations**: Covered ODE/PDE numerics, Monte Carlo methods, and optimization. (Fall 2019)
  - **Finite Mathematics and Programming**: MATLAB programming, logic, and combinatorics. (Spring 2019)
  - **Calculus 2**: Multivariate calculus. (Fall 2018)
  - **Numerical Analysis**: Solving linear systems, power method, numerical integration/differentiation. (Spring 2018)
  - Graded assignments and held office hours across all courses.

Leadership Experience

- **Department seminar organizer as a part of Graduate Student Advisory Committee** *Sep 2022 - May 2025*
  Department of Mathematics and Statistics, University of Massachusetts - Amherst
  - **Invited speakers** among faculties in the department for introducing their research interests to early-career graduate students and **hosted talks**.
  - **Participated in Graduate Student Advisory Committees meetings**, reported the progress, and discussed future directions.

- **Graduate Student Panelist in Open House for Accepted Students** *April 2025*
  Department of Mathematics and Statistics, University of Massachusetts - Amherst
  - **Represented the graduate student body** by sharing perspectives on academic life, research, and department culture with newly admitted PhD and MS students.

- **Panelist in SIAM Graduate School Panel** *Oct 2024*
  Department of Mathematics and Statistics, University of Massachusetts - Amherst
  - **Led a Q&A panel session** to introduce graduate studies in Mathematics to undergraduate mathematics majors.

- **Graduate student representative in Math Majors Fair** *April 2024*
  Department of Mathematics and Statistics, University of Massachusetts - Amherst
  - **Advised undergraduate mathematics majors** on pursuing graduate studies by sharing insights into graduate life and personal research experiences.

Workshops and Training Programs

- **Industrial Mathematics Academy** *Jun 2018*
  National Institute for Mathematical Sciences, South Korea
  - Presented final results of a group project addressing an industrial problem.
  - Proposed a **Convolutional Neural Network** to classify infected individuals from medical images.
  - Coordinated team efforts and participated in tutorials on **Python data analysis** and **Keras**, and lectures on **matrix-based data analysis** and **linear programming**.
- **Industrial Mathematics Academy** *Dec 2017*
  National Institute for Mathematical Sciences, South Korea
  - Proposed a **Poisson process-based model** for assessing driving safety scores from On-board Diagnostic data.
  - Attended tutorials on the **fundamentals of neural networks**.